



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO COCO TIRUCHENGODE (TAPCMS UNIT II)

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

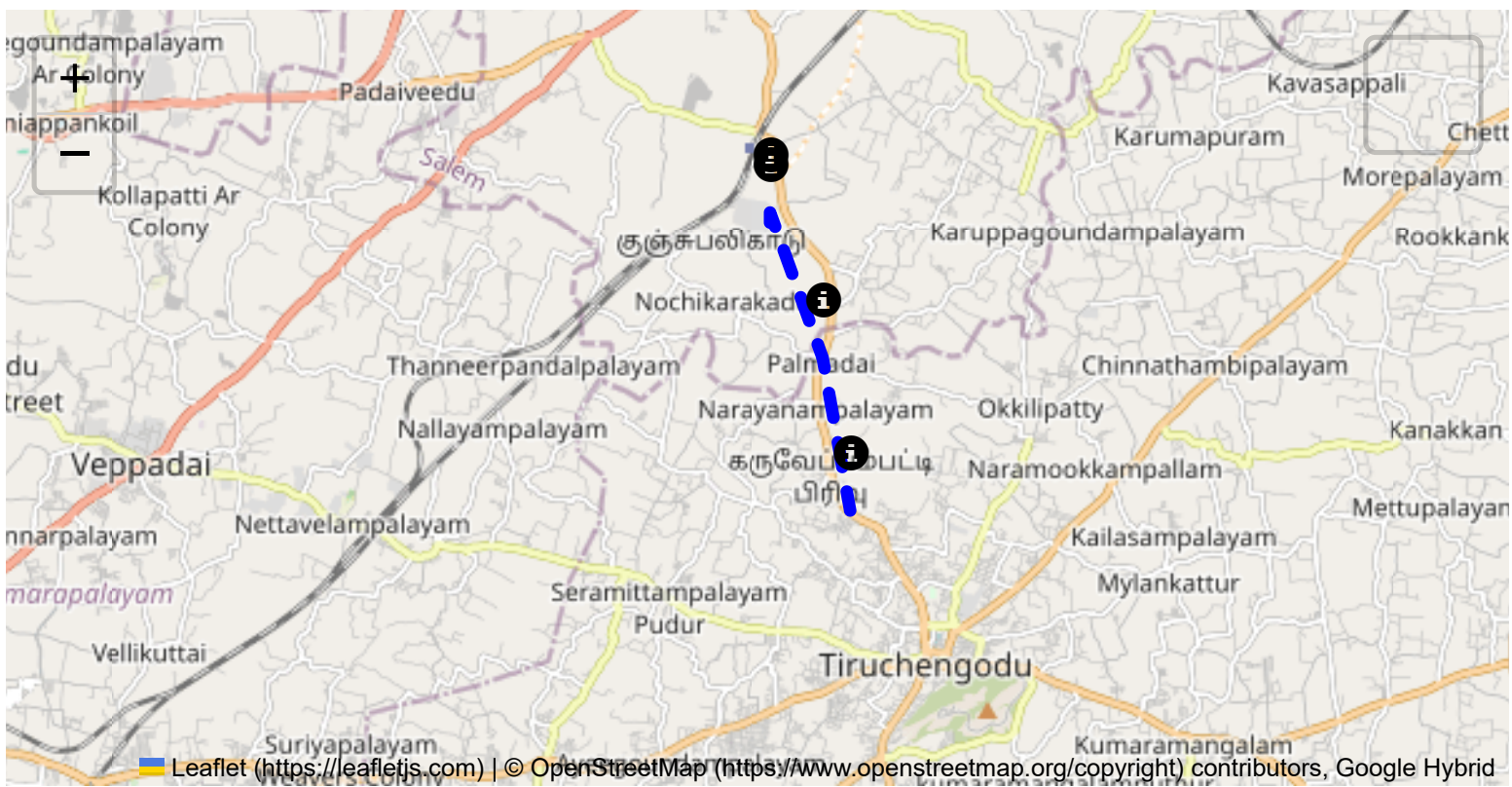
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

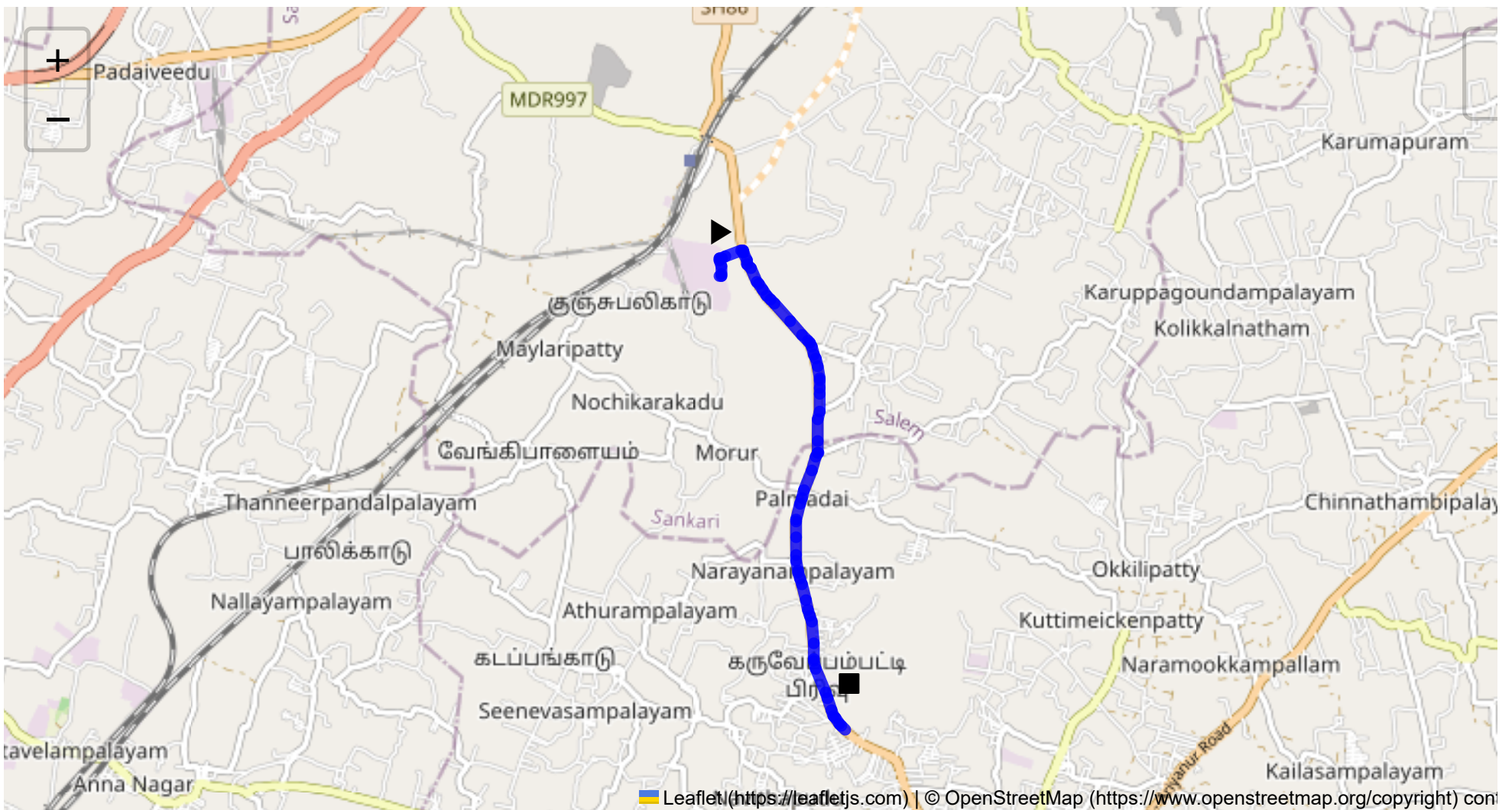
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 5.32 km
Estimated Duration: 0.2 hours
Adjusted Duration (Heavy Vehicle): 0.2 hours
Start: (11.4381, 77.8734)
End: (11.39941, 77.88442)



Welcome to the Journey Risk Management Study

Sure, let's perform an in-depth analysis of the route from CVQF+23W, Sangagiri to 9VXM+RP6, Tiruchengode, via CVQF+VG, Puthur, and CV9J+W7C, SH 86, Palmadai in Tamil Nadu, India. This report focuses on ensuring safety and compliance while transporting hazardous materials over a short 5.32 km distance.

1. Overview of the Route Map

The route spans a rural segment across Tamil Nadu with mostly small roads connecting Sangagiri to Tiruchengode. It traverses through the localities of Puthur and Palmadai. This short route should primarily follow local roads and state highways such as SH 86, providing straightforward navigation due to its brevity.

2. Typical Weather Conditions and Potential Weather-related Hazards

Tamil Nadu experiences a tropical climate, with the monsoon season (June to September) posing significant risks due to heavy rainfall, potentially leading to water-logged patches and reduced visibility. During the summer months (March to May), high temperatures can affect vehicle performance, particularly tire pressure in heavy trucks. Weather-related hazards include:

- Flooding in monsoon season
- Heat stress on vehicles in summer

3. Analysis of Traffic Patterns

The rural nature of the route suggests moderate traffic; however, congestion is still possible around local markets and school zones during the early morning hours and late afternoon/early evening. Congestion typically peaks:

- Between 8–10 AM
- From 4–7 PM

4. Assessment of Road Quality and Infrastructure

Given the short route length, the road quality likely varies from well-maintained state highways to less maintained local roads. Potholes and narrow shoulders could represent safety challenges. The infrastructure may also include limited lighting, making night or low-visibility driving more hazardous.

5. Suggestions for Alternative Routes for Emergencies

Alternative routes would generally align closely along existing rural roads that parallel the primary route; however, each alternate should be scouted beforehand for road conditions. Local advice should be sought in emergencies due to potential limited signage and road quality.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu requires compliance with national hazardous material transport regulations including properly placarded vehicles, adequate documentation, and adherence to designated timings. There may be restrictions on route timings for hazardous materials to avoid school hours and peak traffic.

7. Overview of Historical Incidents

There may not be many publicly documented heavy vehicle or hazardous material incidents on this specific rural route due to its short length and low traffic density. However, routine accident causes, such as vehicle rollovers, may occur due to road conditions or driver error.

8. Environmental Considerations and Sensitive Areas

Watch for agricultural zones and local communities adjacent to the route. Avoid potential spills or emissions that can impact nearby farms, water resources, or cause air contamination. Sensitive cross-border fauna etching zones require extra caution.

9. Analysis of Communication Coverage

This route may have patchy mobile network coverage, particularly in more rural stretches. Prepare by checking coverage maps in advance and confirming availability for emergency communications. Mobile signal issues could create dead zones in more isolated parts of the route.

10. Estimated Emergency Response Times

Response times can vary based on proximity to Sangagiri and Tiruchengode town centers, likely ranging from 15–30 minutes. Outlying areas can experience delayed response due to isolation and road access limitations, especially during inclement weather.

11. Overall Summary of Risk Assessment

- **Traffic & Road Conditions:** Generally manageable, but vigilance required in narrower sections and near populated areas.
- **Weather:** Monsoon poses the greatest hazard; prepare for unexpected conditions.
- **Infrastructure:** Varying road quality and potential communication gaps necessitate careful route planning.
- **Regulation Compliance:** Critical to ensure safe and timely transit of hazardous materials.
- **Emergency Preparedness:** Necessary to rehearse emergency measures due to potential response delays.

Overall, successfully navigating this route for hazardous material transport does involve some risk, particularly regarding road conditions and weather. Pre-trip preparation and on-route vigilance are crucial to safe passage.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43961, 77.87341	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr

Risk Assessment - Elevation Changes

	Risk Type	Specific Risk	Risk Level	Coordinates	Speed Limit
0	Elevation Change	Ascent	Medium	11.40402661250476, 77.88199299460814	30 KMS/Hr
1	Elevation Change	Ascent	Medium	11.4035723760249, 77.88216146772876	30 KMS/Hr

Route Photos of Risky Spots



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544

